

Elixir and Elm – the perfect couple

Tomasz Kowal



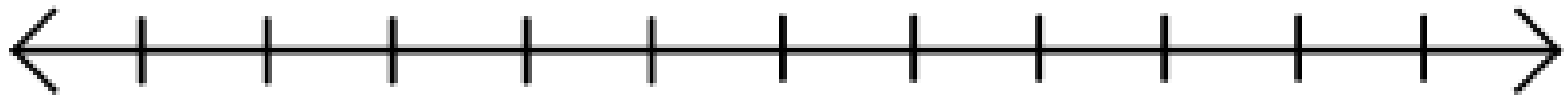
ClubCollect



I hate language comparisons

The Blub paradox

Power continuum





- interactive frontend
- available backend

"it is, all other things being equal, a mistake to
program in anything but the most powerful
[programming language]"

- *Paul Graham*

"it is, all other things being equal, a mistake to
program in anything but the most suitable
language for the problem domain"

- *Tomasz Kowal*

Objective measure

„number of things, you need to keep in mind to
understand given code fragment”
„programmers memory consumption”

PMC

Functional language

```
def function(arg1, arg2) do
  value = some_computations(arg1)
  another_value = some_computations(arg2)
  value + another_value
end
```

Functional language

```
def function(arg1, arg2) do
  value = some_computations(arg1)
  another_value = some_computations(arg2)
  value + another_value
end
```

Functional language

```
def function(arg1, arg2) do  
  value = some_computations(arg1)  
  another_value = some_computations(arg2)  
  value + another_value  
end
```

Functional language

```
def function(arg1, arg2) do
  value = some_computations(arg1)
  another_value = some_computations(arg2)
  value + another_value
end
```

PMC = 4

OO Language

```
def method(arg1, arg2) do
  value = some_computation(arg1 + property1)
  property2 = some_computation(arg2)
  value + property2
end
```

OO Language

property1

property2

def method(arg1, arg2) do

 value = some_computation(arg1 + property1)

 property2 = some_computation(arg2)

 value + property2

end

Building frontend

"modifying document object model by reacting to events from user and server"

Frontend programming

```
$(document).ready(function(){  
    $("p").click(function(){  
        $(this).hide();  
    });  
});
```

Frontend programming

```
$(document).ready(function(){  
    $("p").click(function(){  
        $(this).hide();  
    });  
});
```

Frontend programming

```
$(document).ready(function(){  
    $("p").click(function(){  
        $(this).hide();  
    });  
});
```

Frontend programming

```
$(document).ready(function(){  
    $("p").click(function(){  
        $(this).hide();  
    });  
});
```

Everything that is syntactically legal that the compiler will accept will eventually wind up in your codebase.

- *John Carmack*

Signals



Signals

User input

Server event

User input'

HTML

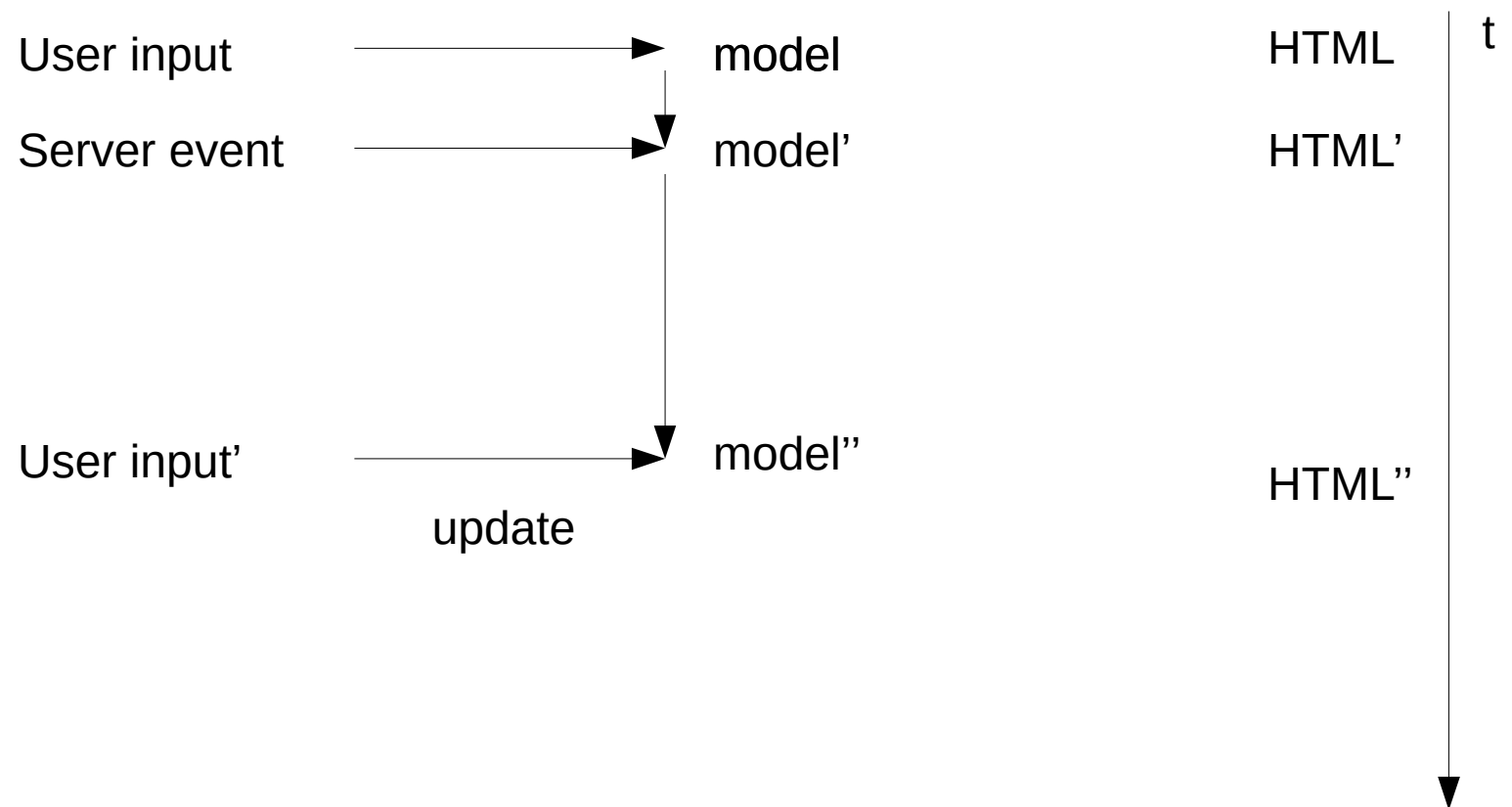
HTML'

HTML''

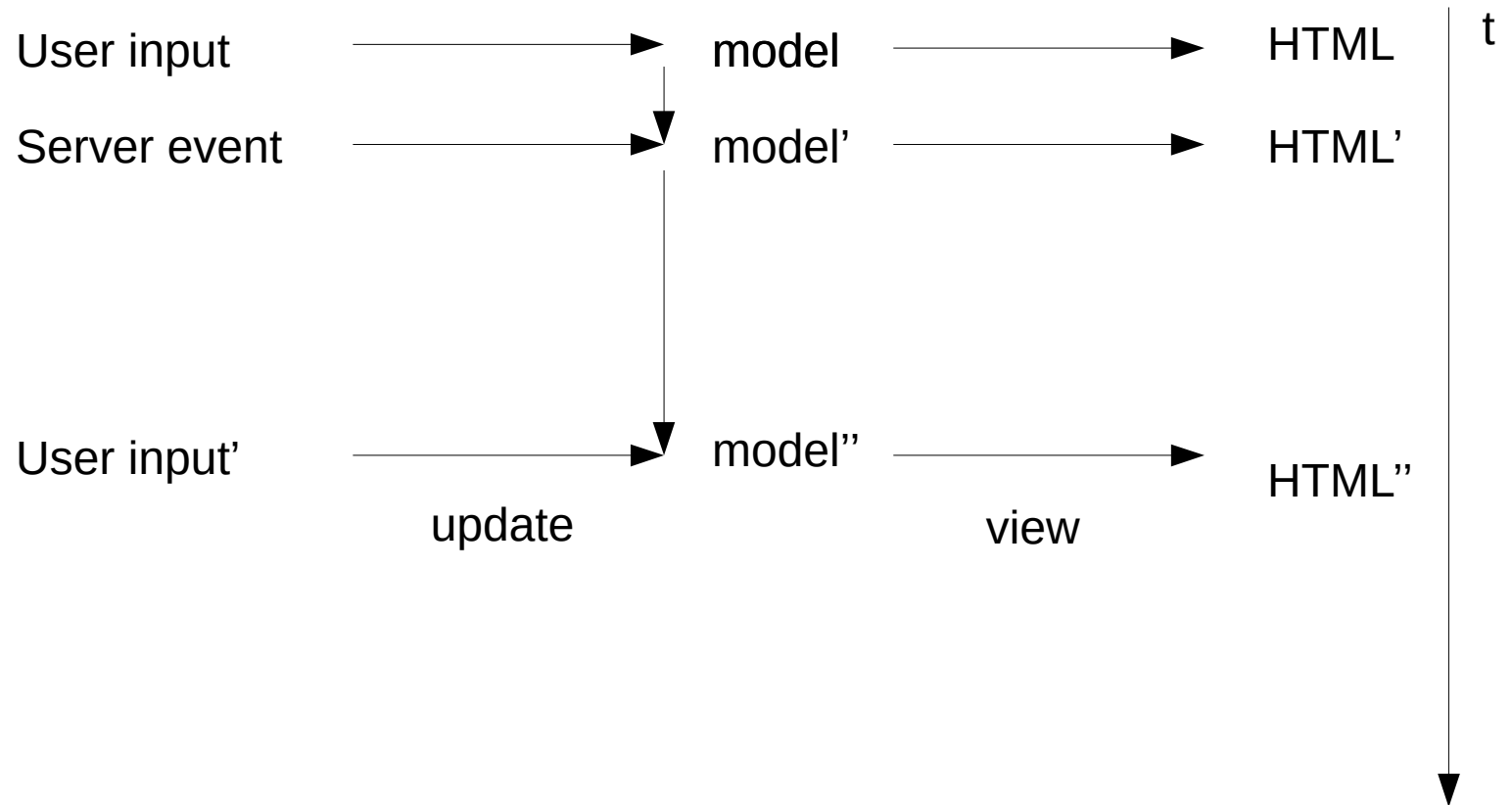
t



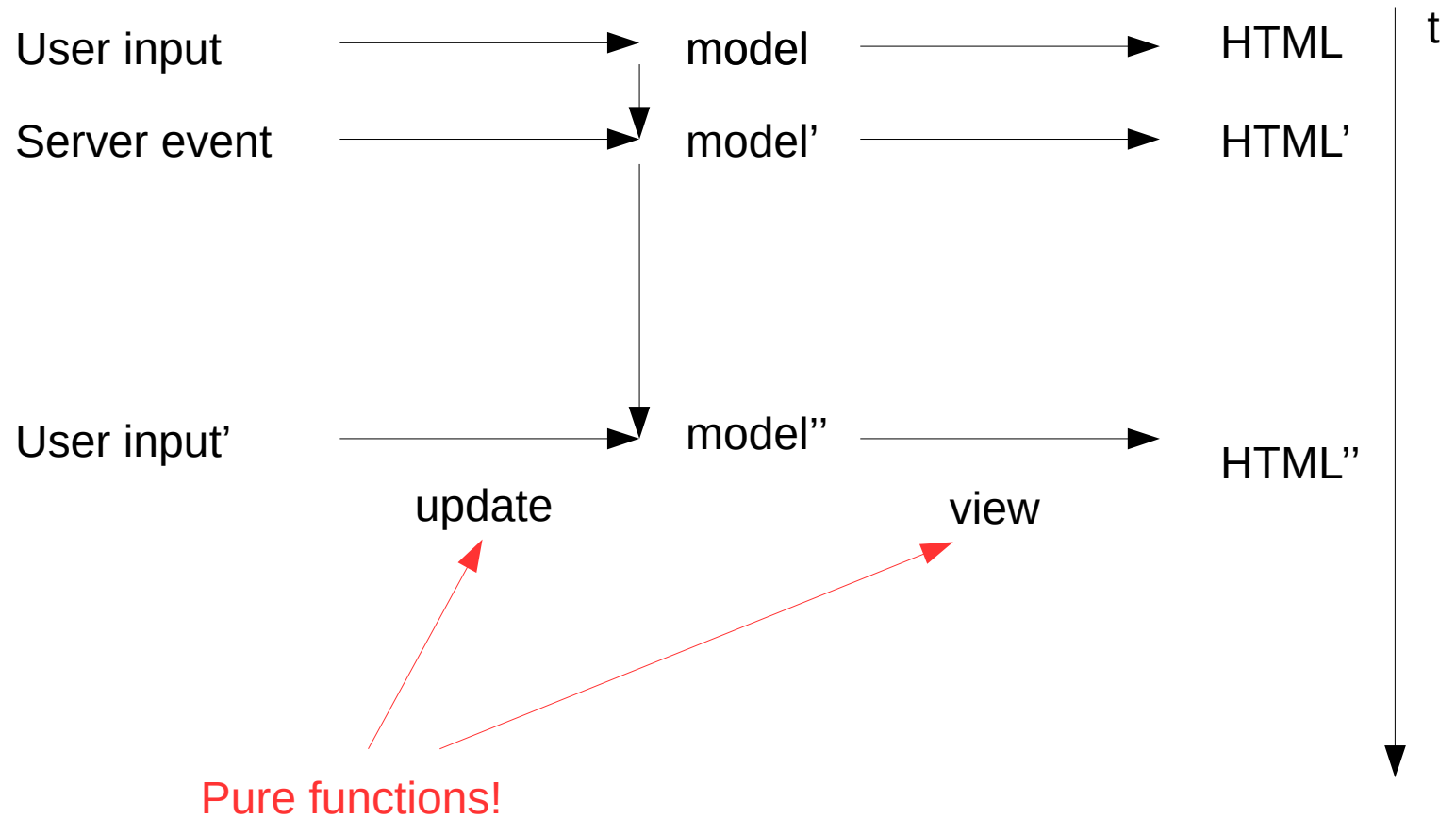
Signals



Signals

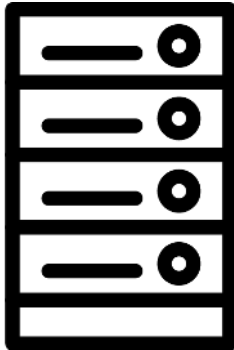


Signals

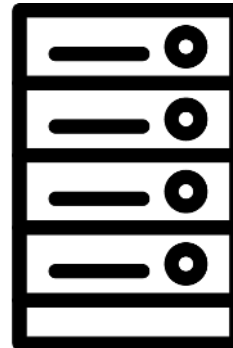
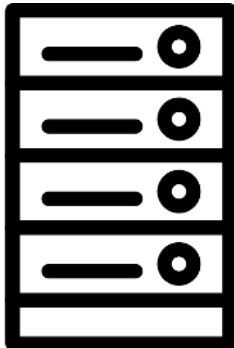
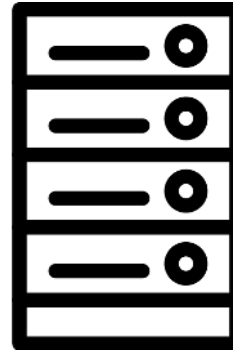
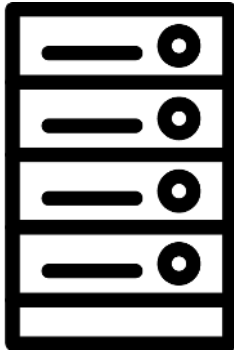


Single data structure holds entire application state.

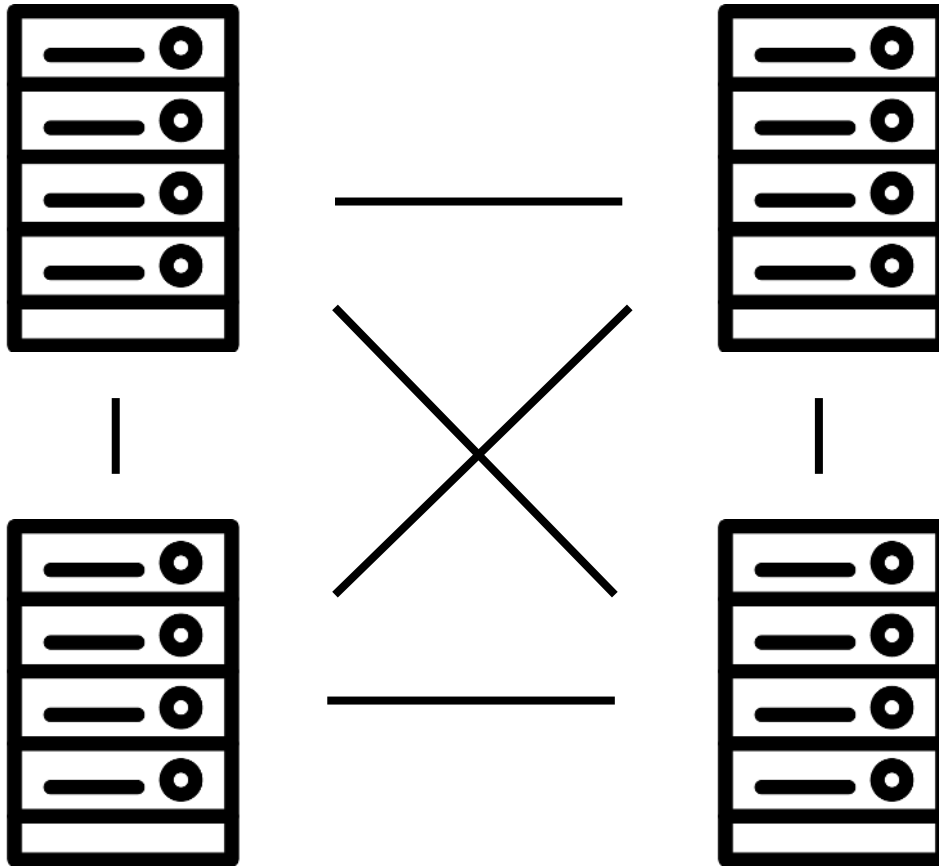
Backend programming



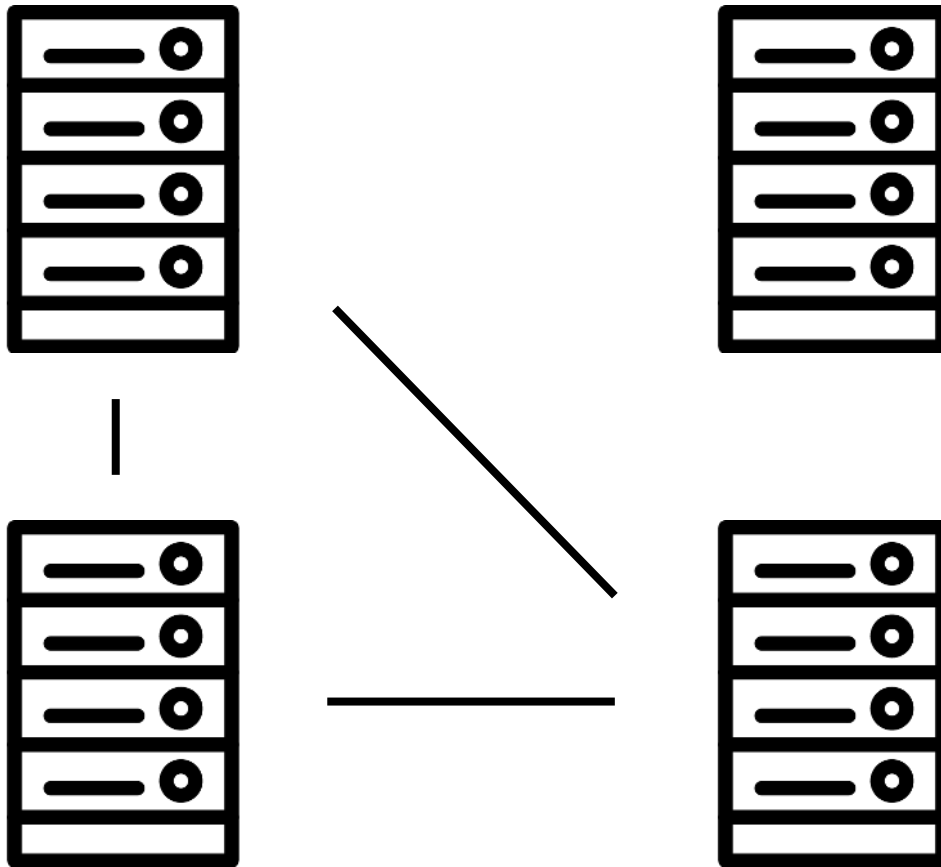
Backend programming



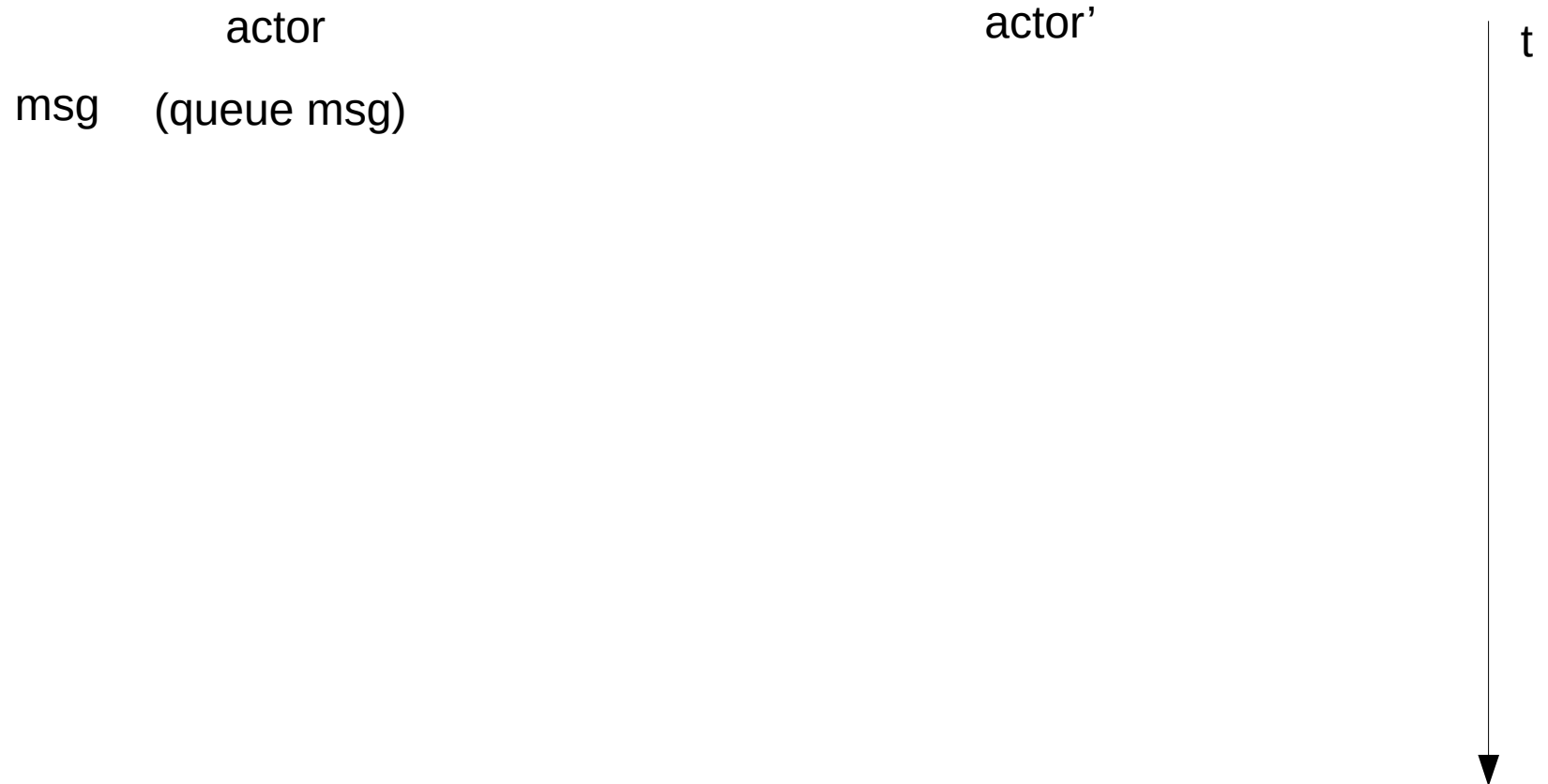
Backend programming



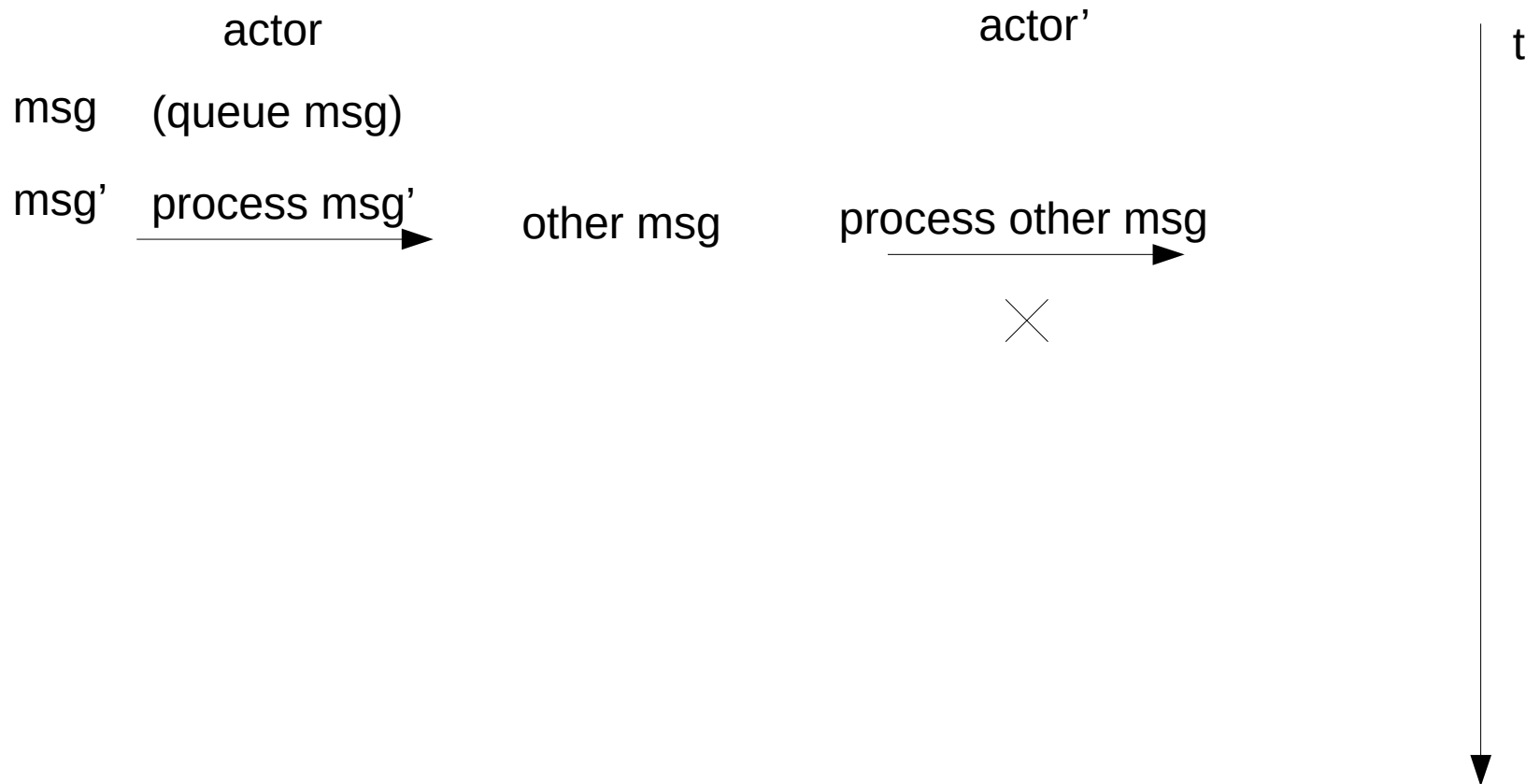
Backend programming - netsplit



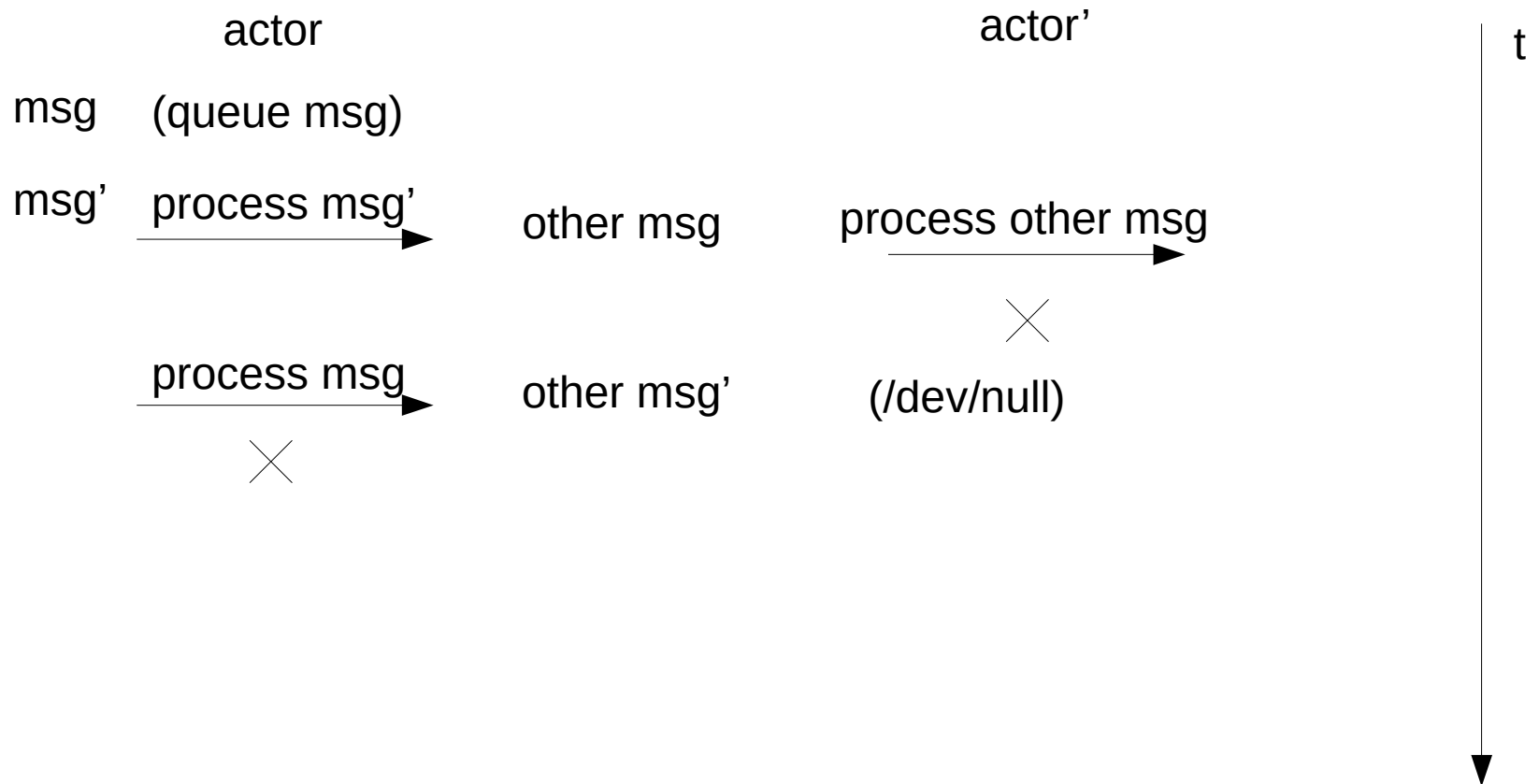
Actors



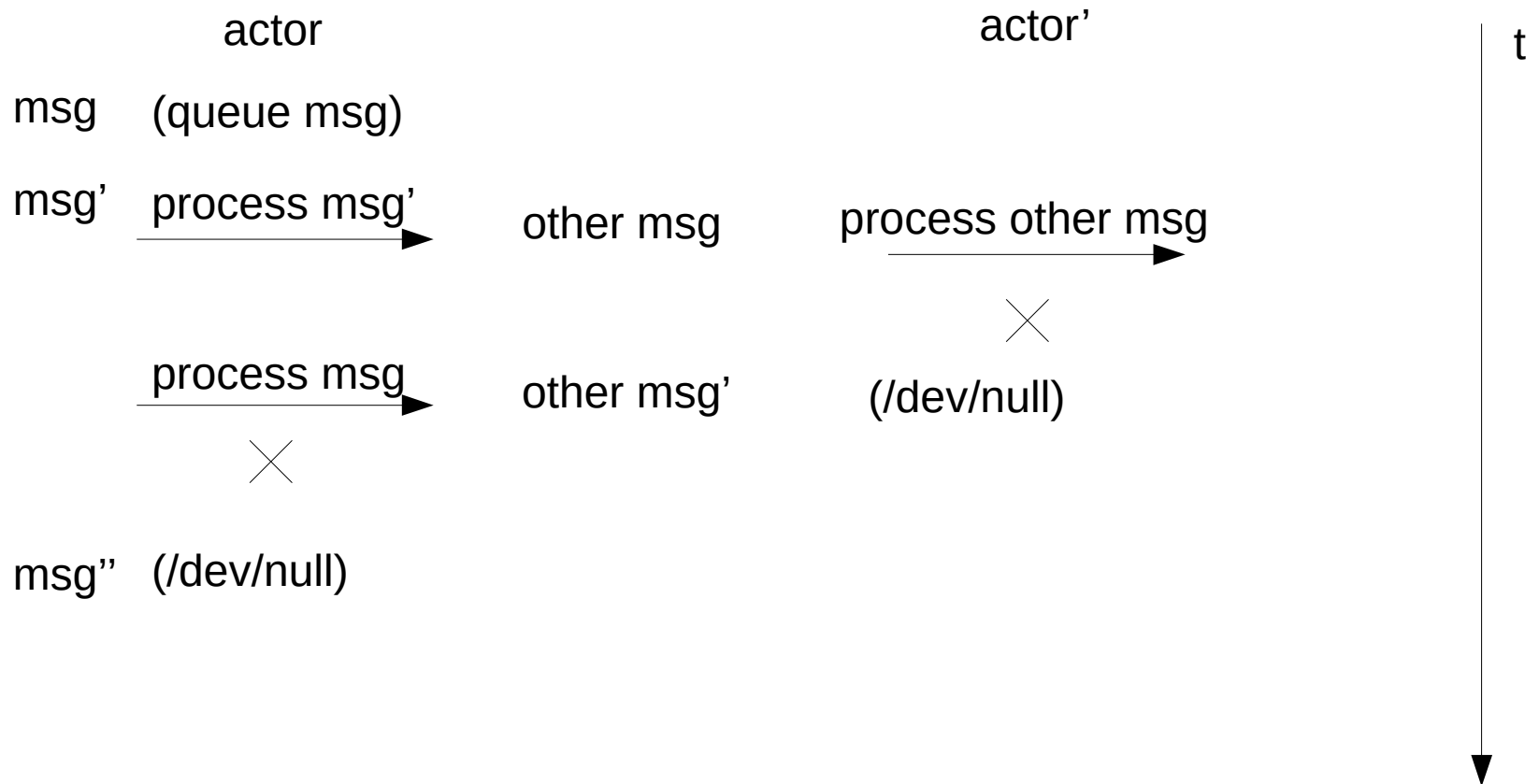
Actors



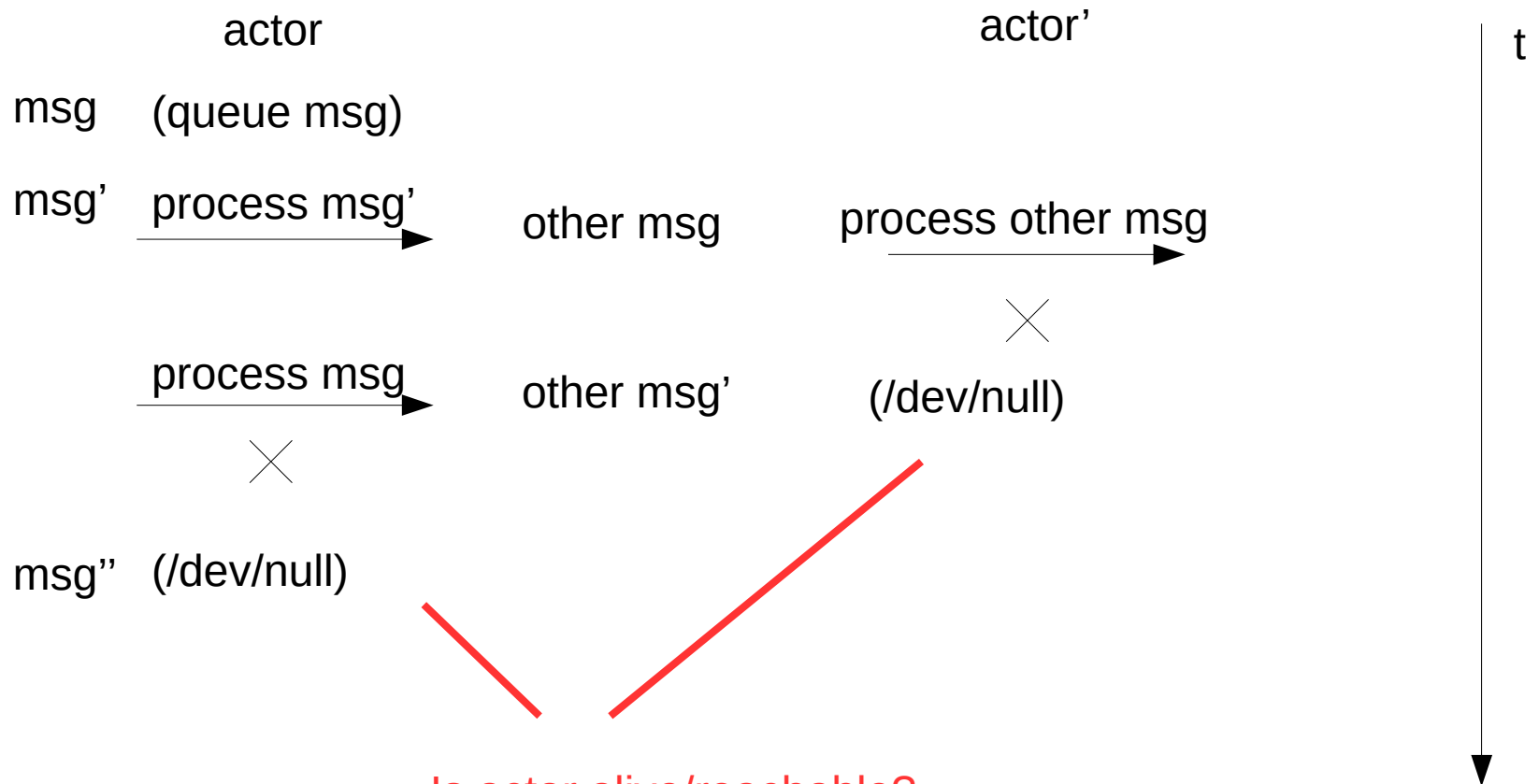
Actors



Actors



Actors



Is actor alive/reachable?
Notify on death
Kill from outside
Kill and restart group of actors

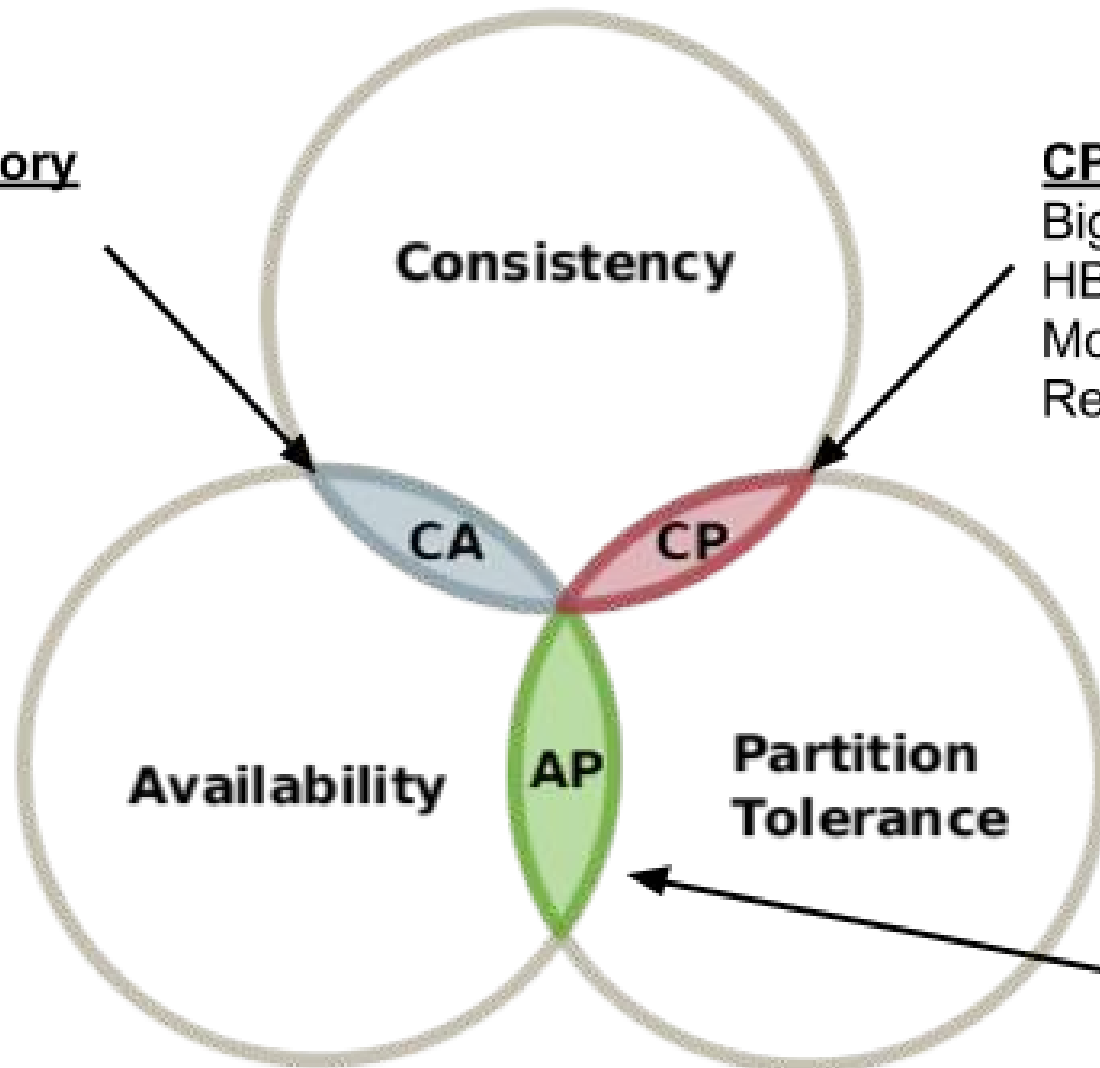
Elixir

```
def function(arg1, arg2) do
  val = receive msg of
    {:msg, msg} -> msg
  end
  val + arg1 + arg2
end
```

CAP

CA Category
RDBMS

CP Category
BigTable
HBase
MongoDB
Redis



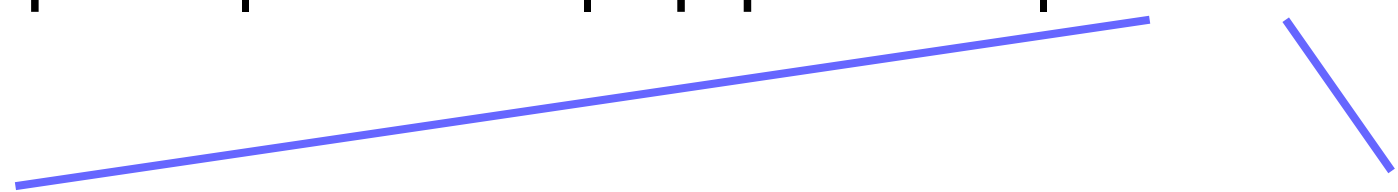
AP Category
Dynamo
Voldemort
Cassandra
CouchDB



Phoenix Framework

conn |> endpoint |> router |> pipelines |> controller

conn |> controller |> common_services |> action



The diagram illustrates two distinct paths from the initial 'conn' object to the 'controller' object. The top path is a linear sequence: 'conn' |> 'endpoint' |> 'router' |> 'pipelines' |> 'controller'. The bottom path is: 'conn' |> 'controller' |> 'common_services' |> 'action'. A blue line connects the 'conn' of the top path to the 'conn' of the bottom path. Another blue line connects the 'controller' of the top path to the 'controller' of the bottom path, indicating that both paths converge at the 'controller' stage.

- Single data structure (source of truth)
 - flow in one direction

Things to remember #1

Different programming languages solve different problems.

Things to remember #2

For interactive web applications check out both
Elixir and Elm

Things to remember #3

If you like this talk, follow me on Twitter
[@snajper47](https://twitter.com/snajper47)

Elm lang homepage: <http://elm-lang.org/>
Elixir lang homepage: <http://elixir-lang.org/>
Phoenix framework:
<http://www.phoenixframework.org/>
My blog: <http://tkowal.wordpress.com/>

Review: Krzysztof Wende